4.5 Part – A : Two Marks Questions

Question 92. Write down the principle of least squares.
Nov.'12, Apr.'13, Apr.'15, Nov.'15

Answer . The best fit of the curve is that for which the residual error at each point of x_i for the given pairs (x_i, y_i) are as small as possible. Then the sum of squares of the errors is a minimum. This is known as the principle of least squares.

Question 93. Define curve fitting.

Answer . Curve fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points, possibly subject to constraints. (or)

Curve fitting is a mathematical model described by equations or systems of equations to observations in order to estimate parameters that can be used to interpret the experimental data.

Question 94. What is the error committed, when fitting a straight line of the form $y = ax + b$ for the observed values (x_i, y_i) : $i = 1, 2, 3, ...n$. **Apr.'12**

Answer . When we fit a straight line $y = ax + b$ for given observed values (x_i, y_i) ; $i = 1, 2, ...n$ then $y - y_i$ is a error called residual error.

Question 95. Write the normal equations to fit a curve the straight line $y = ax + b$ by the method of least squares.

Solution . The normal equations to fit a curve in the straight line $y = ax + b$ is

$$\begin{aligned} a\Sigma x + nb &= \Sigma y \\ a\Sigma x^2 + b\Sigma x &= \Sigma xy \end{aligned}$$

Question 96. Write down the normal equations to fit the parabola $y = ax^2 + bx + c$ for the method of least squares. **Apr.'11, Nov.'15**

Answer . The normal equations of the parabolic fit of the curve are

$$\begin{aligned} a\Sigma x^2 + b\Sigma x + nc &= \Sigma y \\ a\Sigma x^3 + b\Sigma x^2 + c\Sigma x &= \Sigma xy \\ a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2 &= \Sigma x^2 y \end{aligned}$$

Question 97. Write the normal equations to fit a curve of the form $y = ax^b$ by the method of least squares. **Nov.'11**

Answer . The given fit of the curve is $y = ax^b$. ...(1)

Taking logarithm on both sides of (1), we get

$$\begin{aligned} \log_{10} y &= \log_{10}\left(ax^b\right) \\ &= \log_{10} a + \log_{10} x^b \\ \therefore \quad \log_{10} y &= \log_{10} a + b\log_{10} x \\ \text{i.e.,} \quad Y &= A + bX \qquad \text{...(2)} \end{aligned}$$

Then we determine the parameters A and B by using method of least squares. The normal equations are

$$\begin{aligned} nA + b\Sigma X &= \Sigma Y \\ A\Sigma X + b\Sigma X^2 &= \Sigma XY \end{aligned}$$

Question 98. Define 'Population'. **Nov.'14**

Solution . A population consists of collection of individuals, which may be person or experimental outcomes, whose characteristic are to be studied.

Question 99. Define sample.

Solution . A sample is a portion of the population that is studied to learn about the characteristics of the population.

Question 100. Define 'Large and Small samples.

Nov.13, Nov.'14, Nov.'15

Solution . If the number of elements (or) items (say n) in the sample is greater than or equal to 30 (i.e., $n \geq 30$), then the sample is called as large samples. If the number of elements (or) items (n) in the sample is less than 30 (i.e.,$n < 30$), then it is called as small samples.

Question 101. Define a residual. **Nov.'13**

Solution . The residual (of an observed value) is the difference between the observed value and the estimated function value.

Question 102. Define standard error.

Solution . The standard deviation of the sampling distribution of a statistic is called standard error. Standard error is used to access the difference between the expected and observed values.

Question 103. What is the formula for finding the difference between the sample proportion and population proportion? **Apr.'13**

Answer . Let P is the population proportion and p is the sample proportion then

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \text{ and } Q = 1 - P$$

Question 104. Define hypothesis.

Solution . A hypothesis is a statement of a relationship between two or more variables. A statistical hypothesis is simply a particular kind of hypothesis.

Question 105. Define Statistical hypothesis.

Solution . A statistical hypothesis is either

1. A statement about the value of a population parameter (e.g., mean, median, mode, variance, standard deviation etc), or

2. A statement about the kind of probability distribution that a certain variable obeys.

Question 106. Define "test statistic."

Solution . The test statistic is a mathematical formula to determine the likelihood of obtaining sample outcomes if the null hypothesis were true. The value of the test statistic is used to make a decision regarding the null hypothesis.

Question 107. What are the null and alternative hypothesis?
Nov.'10, Nov.'11, Apr.'12, Apr.'14

Solution . Null hypothesis always predicts that there is no relationship between the variables being studied and it is denoted by H_0. The null hypothesis typically corresponds to a general or a default position. Making this assertion will make no difference and hence cannot be proven positively.

An alternative hypothesis (H_1) is a statement that directly contradicts a null hypothesis by stating that that the actual value of a population parameter is less than, greater than, or not equal to the value stated in the null hypothesis.

Question 108. Define parameters and statistics.

Solution . Parameters are the measures such as mean, standard deviation etc which describe a population and the word statistic is used is used to indicate various measures relating to the sample such as mean, S.D., etc. A statistic is any function of observations in a random sample.

Question 109. Define sampling distribution

Solution . The probability distribution of a statistic is called as a sampling distribution

Question 110. Define critical region or region of rejection.

Solution . A region corresponding to a statistic, in the sample space which amounts to rejection of null hypothesis H_0 is called as critical region. The region of the sample space which amounts to the acceptance of null hypothesis is called acceptance region.

Question 111. Define critical value (or) significant value.

Solution . The value of the test statistic which separates the acceptance region and critical region is called the critical value.

Question 112. What is one tailed test? **Apr.'14**

Solution . A one-tailed test uses an alternate hypothesis that states either $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$, but not both. One tailed test is also classified as (i) right tailed test (ii) left tailed test.

Question 113. Define right-tail test and left-tail test.

Solution . If the observed test values fall to the right of the critical value then the test is called as right-tail test. In a right-tail test, we accept the null hypothesis if the observed test value is less than or equal to the critical value.

It the observed test values fall to the left of the critical value then the test is called as left-tail test. In a left-tail test, we accept the null hypothesis if the observed test value is greater than or equal to the critical value.

Question 114. Define two tailed test.

Solution . When two tails of the sampling distribution of the normal curve are used, then the test is called as two-tailed test. In the two tailed test the alternative hypothesis is stated as $\neq$.

Question 115. Define Test of significance. **Apr.'15**

Solution . Significance testing (also called hypothesis testing) is a mathematical method which is used to decide whether or not an hypothesis (an assumption made about a population parameter) should be accepted or rejected based upon our sample statistic.

Question 116. Define 'Level of Significance'. **Apr.'11, Nov.'15**

Solution . The level of significance is the maximum probability of committing a type I error. This probability is symbolized by α. That is, P(type I error) $= \alpha$. and P(type II error) $= \beta$.

It is the probability level below which the null hypothesis is rejected. Generally 5% and 1% level of significance are used.

Question 117. What are the Type I and Type II errors in sampling theory? **Nov.'11, Apr.'11, Nov.'15**

Solution . Type I error is the probability of rejecting a null hypothesis (H_0) when it is actually true. Type II error is the probability of accepting a null hypothesis (H_0) when it is false.

Question 118. A sample of 900 items has mean 3.4 and standard deviation 2.61. Can the sample regarded as drawn from a population with mean 3.25 at 5% level of significance? **Apr.'11**

Solution . Given $n = 900$, $\overline{x} = 3.4$, $\mu = 3.25$ and $\sigma = 2.61$
Null hypothesis (H_0) : $\mu = 3.25$.
Alternative hypothesis (H_1) : $\mu \neq 3.25$.

$$\therefore\ z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$$

Tabulated value of z at 5% level of significance $= 1.96$
At 5% level, the calculated value of $z <$ tabulated value of z.
$\therefore$ Accept H_0. Difference is not significant.

Question 119. A sample of 900 members from a normal population with SD 2.61 cms has a mean 3.5 cms. Find the 95% fiducial limits for the population mean. **Nov.'10**

Solution . Given $n = 900$, $\overline{x} = 3.5$ and $\sigma = 2.61$

If the population mean μ is unknown, 95% confidence limits of μ are

$$= \overline{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 3.5 \pm 1.96 \times \frac{2.61}{\sqrt{900}}$$

$$= 3.5 \pm 0.17 = (3.33,\ 3.67)$$

Question 120. Write down the test statistic for large sample test for difference of means. **Nov.'11**

Solution . If the samples are drawn from the same population with known S.D σ then the test statistic is

$$z = \frac{\overline{x_1} - \overline{x_2}}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Question 121. A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased **Nov.'12**

Solution . Given that, sample size $n = 400$. Number of heads $= 216$.

$p =$ sample proportion of heads $= \frac{216}{400} = 0.54$

$P =$ Population proportion $= \frac{1}{2} = 0.5$. $\therefore$ $Q = 1 - P = 0.5$

Null hypothesis (H_0) : Coin is unbiased. or $P = \frac{1}{2}$

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$.

$$\therefore z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} = 1.6$$

$\therefore$ The calculated value of $z = 1.6$

Tabulated value of z at 5% level of significance $= 1.96$

Since calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Thus coin is unbiased.